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Practical No. 9

Aim: To implement the K-Means clustering algorithm and analyze the clustering results using appropriate evaluation techniques.

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<u>Aim:</u> To implement the K-Means clustering algorithm and analyze the clustering results using appropriate evaluation techniques.

<u>OBJECTIVE/EXPECTED LEARNING OUTCOME:</u>

The objectives and expected learning outcome of this practical are:

- To understand the working of the K-Means clustering algorithm.
- To determine the optimal number of clusters using methods such as the Elbow Method.
- To apply K-Means clustering to segment customers.
- To evaluate clustering quality using metrics such as Inertia and Silhouette Score.

THEORY:

Objective of k means Clustering:

The main objective of k-means clustering is to partition your data into a specific number (k) of groups, where data points within each group are similar and dissimilar to points in other groups. It achieves this by minimizing the distance between data points and their assigned cluster‟s center, called the centroid. Here‟s an objective:

- **Grouping similar data points:** K-means aims to identify patterns in your data by grouping data points that share similar characteristics together. This allows you to discover underlying structures within the data.
- **Minimizing within-cluster distance:** The algorithm strives to make sure data points within a cluster are as close as possible to each other, as measured by a distance metric (usually Euclidean distance). This ensures tight-knit clusters with high cohesiveness.
- **Maximizing between-cluster distance:** Conversely, k-means also tries to maximize the separation between clusters. Ideally, data points from different clusters should be far apart, making the clusters distinct from each other.

What is Clustering?

Cluster analysis is a technique in **data mining** and machine learning that groups similar objects into clusters. K-means clustering, a popular method, aims to divide a set of objects into K clusters, minimizing the sum of squared distances between the objects and their respective cluster centers. Hierarchical clustering and k-means clustering are two popular techniques in the field of unsupervised learning used for clustering data points into distinct groups. While k-means

clustering divides data into a predefined number of clusters, hierarchical clustering creates a hierarchical tree-like structure to represent the relationships between the clusters. Example of Clustering

Let‟s try understanding this with a simple example. A bank wants to give credit card offers to its customers. Currently, they look at the details of each customer and, based on this information, decide which offer should be given to which customer.

Now, the bank can potentially have millions of customers. Does it make sense to look at the details of each customer separately and then make a decision? Certainly not! It is a manual process and will take a huge amount of time.

So what can the bank do? One option is to segment its customers into different groups. For instance, the bank can group the customers based on their income:

Can you see where I‟m going with this? The bank can now make three different strategies or offers, one for each group. Here, instead of creating different strategies for individual customers, they only have to make 3 strategies. This will reduce the effort as well as the time.

What is the Elbow Method?

The Elbow Method is a technique used in data analysis and **machine learning** for determining the optimal number of clusters in a dataset. It involves plotting the variance explained by different numbers of clusters and identifying the "elbow" point, where the rate of variance decreases sharply levels off, suggesting an appropriate cluster count for analysis or model training.

This method is a visual technique used to determine the best K value for a k-means clustering algorithm. In this method, a graph known as the elbow graph plots the withincluster-sum-of-square (WCSS) values against various K values. The optimal K value is identified at the point where the graph bends like an elbow.

What is the Elbow Method in K-Means Clustering?

The elbow method is a graphical representation of finding the optimal „K‟ in a K-means clustering. It works by finding WCSS (Within-Cluster Sum of Square) i.e. the sum of the square distance between points in a cluster and the cluster centroid.

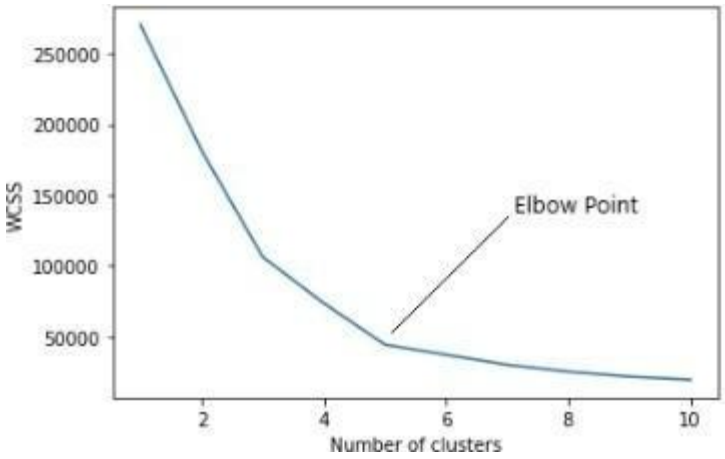
Let‟s go through the steps involved in K-means clustering for a better understanding:

- Select the number of clusters for the dataset (K)
- Select the K number of centroids randomly from the dataset.
- Now we will use Euclidean distance or Manhattan distance as the metric to calculate the distance of the points from the nearest centroid and assign the points to that nearest cluster centroid, thus creating K clusters.
- Now we find the new centroid of the clusters thus formed.
- Again reassign the whole data point based on this new centroid, then repeat step 4. We will continue this for a given number of iterations until the position of the centroid doesn‟t change, i.e., there is no more convergence.

Finding the optimal number of clusters is an important part of this algorithm. A commonly used method for finding the optimum K value is **Elbow Method.**

K Means Clustering Using the Elbow Method:

In the Elbow method, we are actually varying the number of clusters (K) from 1 – 10. For each value of K, we are calculating WCSS (Within-Cluster Sum of Square). WCSS is the sum of the squared distance between each point and the centroid in a cluster. When we plot the WCSS with the K value, the plot looks like an Elbow. As the number of clusters increases, the WCSS value will start to decrease. WCSS value is largest when K = 1. When we analyze the graph, we can see that the graph will rapidly change at a point and thus creating an elbow shape. From this point, the graph moves almost parallel to the X-axis. The K value corresponding to this point is the optimal value of K or an optimal number of clusters.



Silhouette Score:

The silhouette score and plot are used to evaluate the quality of a clustering solution produced by the k-means algorithm. The silhouette score measures the similarity of each point to its own cluster compared to other clusters, and the silhouette plot visualizes these scores for each sample. A high silhouette score indicates that the clusters are well separated, and each sample is more similar to

the samples in its own cluster than to samples in other clusters. A silhouette score close to 0 suggests overlapping clusters, and a negative score suggests poor clustering solutions.

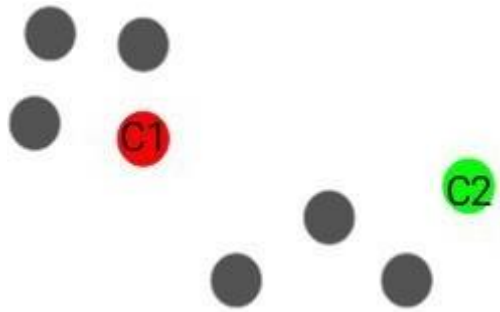
How to Apply K-Means Clustering Algorithm?

Let‟s now take an example to understand how K-Means Clustering actually works:

We have these 8 points, and we want to apply k-means to create clusters for these points.
Here‟s how we can do it.

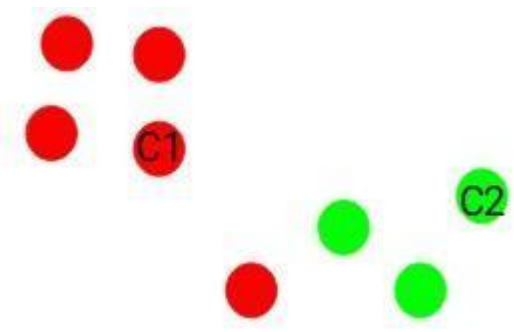
1. **Choose the number of clusters *k***
   The first step in k-means is to pick the number of clusters, k.

2. **Select k random points from the data as centroids** Next, we randomly select the centroid for each cluster. Let‟s say we want to have 2 clusters, so k is equal to 2 here. We then randomly select the centroid:



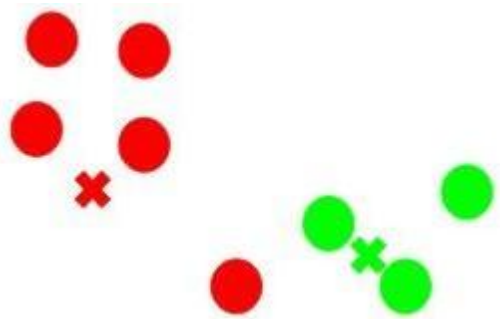
   Here, the red and green circles represent the centroid for these clusters.

3. **Assign all the points to the closest cluster centroid**
   Once we have initialized the centroids, we assign each point to the closest cluster centroid:



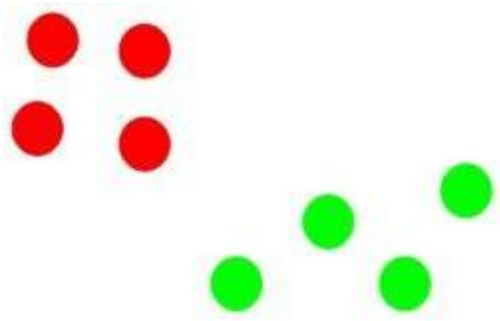
Here you can see that the points closer to the red point are assigned to the red cluster, whereas the points closer to the green point are assigned to the green cluster.

4. **Recompute the centroids of newly formed clusters**
   Now, once we have assigned all of the points to either cluster, the next step is to compute the centroids of newly formed clusters:



Here, the red and green crosses are the new centroids.

5. **Repeat steps 3 and 4** We then repeat steps 3 and 4:



The step of computing the centroid and assigning all the points to the cluster based on their distance from the centroid is a single iteration. But wait – when should we stop this process? It can‟t run till eternity, right?

Stopping Criteria for K-Means Clustering

There are essentially three stopping criteria that can be adopted to stop the K-means algorithm:

1. **Centroids of newly formed clusters do not change**
2. **Points remain in the same cluster**
3. **Maximum number of iterations is reached**

We can stop the algorithm if the centroids of newly formed clusters are not changing. Even after multiple iterations, if we are getting the same centroids for all the clusters, we can say that the algorithm is not learning any new pattern, and it is a sign to stop the training. Another clear sign that we should stop the training process is if the points remain in the same cluster even after training the algorithm for multiple iterations.

Finally, we can stop the training if the maximum number of iterations is reached. Suppose we have set the number of iterations as 100. The process will repeat for 100 iterations before stopping.

Advantages of K-Means

1. Customer segmentation
2. Image segmentation
3. Market analysis
4. Anomaly detection
5. Document clustering

Advantages of K-Means

1. Simple and easy to implement
2. Computationally efficient and fast
3. Scales well to large datasets
4. Works well for well-separated clusters
5. Easy to interpret results

Disadvantages of K-Means:

1. Requires predefining number of clusters (K)
2. Sensitive to initial centroid selection
3. Not suitable for non-spherical clusters
4. Sensitive to outliers and noise
5. Performance depends on feature scaling

PROGRAM CODE:

CONCLUSION:

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DISCUSSION AND VIVA VOCE:

1. How would you use K-Means to segment customers in a retail business, and how would you decide the optimal number of clusters?
2. In a banking dataset with 50+ features, how would you select the right features before applying K-Means?
3. Why is feature scaling important in K-Means, and what could go wrong if you skip it in a real-world application?
4. Apart from the Elbow Method, what other techniques would you use to determine the optimal number of clusters in production?
5. How would you implement K-Means efficiently on a dataset with millions of records?
6. How do outliers affect K-Means clustering, and what strategies would you use to handle them?
7. Once clusters are formed, how would you interpret and label them for business stakeholders?
8. In a real-world scenario where true labels are not available, how would you evaluate the quality of clustering?
9. Can you describe a business scenario where K-Means would fail and another algorithm (like DBSCAN or Hierarchical clustering) would be better?
10. How would you design a system that uses K-Means for real-time customer segmentation or recommendation?

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